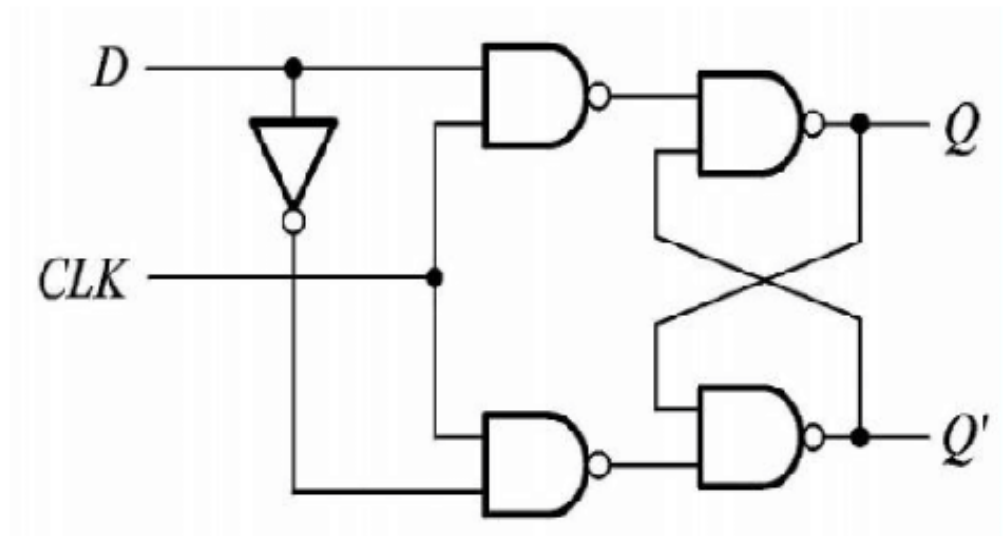


Day 28

D Flip-Flop (Edge-Triggered with Reset) in Verilog

Circuit Diagram :



Q	D	$Q(t+1)$
0	0	0
0	1	1
1	0	0
1	1	1

Verilog Code

D-FlipFlop verilog code

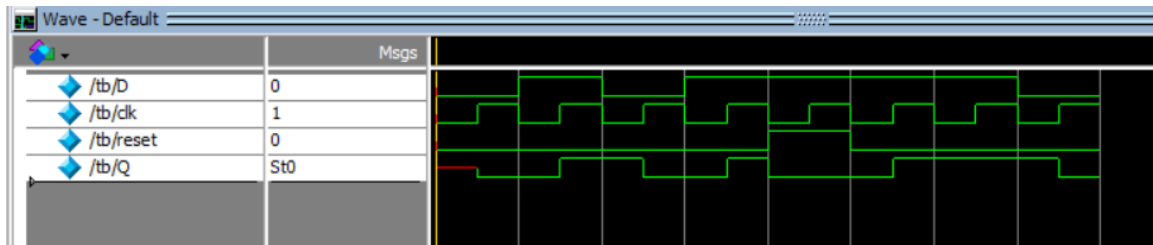
```
1  module D_FF(D,clk,reset,Q);
2  input D,clk,reset;
3  output reg Q;
4  always @(posedge clk or posedge reset) begin
5  if (reset)
6  Q <= 0;
7  else
8  Q <= D;
9  end
10 endmodule
```

Testbench Code

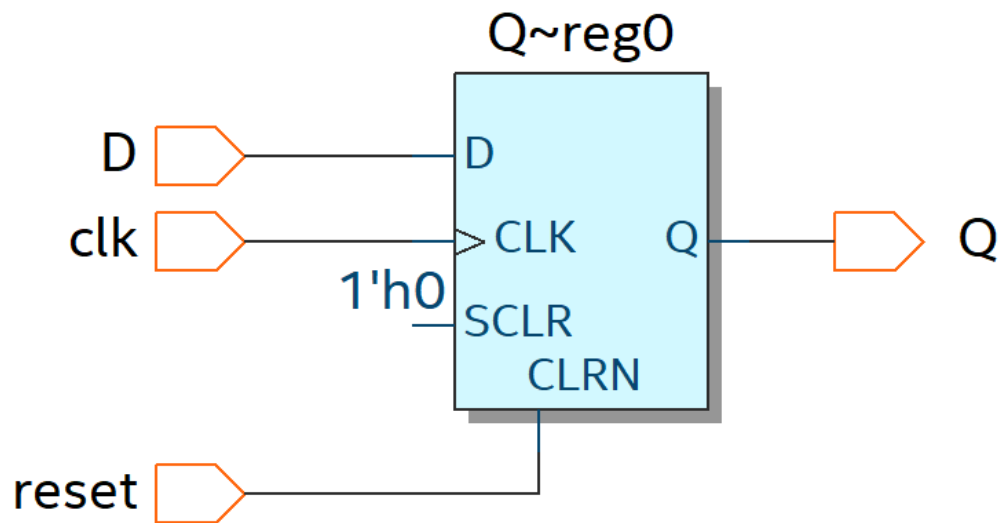
D-FlipFlop Tb code

```
1  module tb;
2  reg D,clk,reset;
3  wire Q;
4  D_FF dut(D,clk,reset,Q);
5  initial
6  begin
7  clk=0;
8  forever #5 clk=~clk;
9  end
10 initial
11 begin
12 reset=0;
13 D=0;#10
14 D=1;#10
15 D=0;#10
16 D=1;#10
17 reset=1;#10
18 reset=0;#10
19 D=1;#10
20 D=0;#10
21 $finish;
22 end
23 endmodule
```

Waveform



Schematic



EDA Tools Used

- IntelQuartusPrime
- ModelSim